Statistical Methods in Chemistry

ANOVA in Chemistry

Comparing Three or More Experimental Means

Student worksheet

Name: ______________________________

Class: ______________________________

Date: ______________________________

Contents

1 Why ANOVA appears in chemistry

Chemists often compare more than two experimental conditions. For example, a class may compare three analytical methods for the same ion, three catalysts for the same reaction, four water sources, or several temperatures in a rate investigation.

A two-sample t-test is designed for two means. If there are three or more means, repeatedly applying t-tests is a poor habit because each test gives another chance of a false positive. Analysis of variance, abbreviated **ANOVA**, tests the whole set of means in one controlled comparison.

Core question

One-way ANOVA asks:

Are the observed differences between group means large compared with the scatter inside the groups?

If the group means are far apart but each group is internally tight, the evidence for a real difference is stronger. If the groups are internally scattered, differences between means may be just experimental noise.

Situation	Better statistical tool
Compare one mean with a known certified value	One-sample t-test
Compare two independent means	Two-sample t-test
Compare before/after readings on the same samples	Paired t-test
Compare three or more group means	One-way ANOVA
Compare two factors at once, such as catalyst and temperature	Two-way ANOVA

Quick exercise: Choosing the method

For each chemistry situation, write whether ANOVA is appropriate.

a) Three brands of orange juice are compared for vitamin C concentration using iodometric titration.

b) A student's measured calcium concentration is compared with a certified value of $100.0\,\mathrm{mg\,L^{-1}}$.

c) Two indicators are compared for the same acid-base titration.

d) Four catalysts are compared by measuring percentage yield in the same reaction.

2 The one-way ANOVA model

Suppose there are k groups. Group i contains n_i measurements. A single measurement is written as x_{ij}, where i identifies the group and j identifies the replicate inside that group.

Means used in ANOVA

$$\bar{x}_i = \text{mean of group } i$$

$$\bar{x}_{\text{grand}} = \text{mean of all measurements combined}$$

ANOVA compares how far the group means are from the grand mean with how far individual readings are from their own group mean.

Two sources of variation

$$SS_{\text{between}} = \sum_i n_i(\bar{x}_i - \bar{x}_{\text{grand}})^2$$

$$SS_{\text{within}} = \sum_i \sum_j (x_{ij} - \bar{x}_i)^2$$

$$SS_{\text{total}} = \sum_i \sum_j (x_{ij} - \bar{x}_{\text{grand}})^2 = SS_{\text{between}} + SS_{\text{within}}$$

Term	Meaning
SS_{between}	variation explained by group membership; the means are separated
SS_{within}	unexplained scatter inside groups; ordinary experimental variability
SS_{total}	total variation in all measurements

The ANOVA table

Source	Sum of squares	Degrees of freedom	Mean square	Test statistic
Between groups	SS_{between}	$k-1$	$MS_{\text{between}} = \dfrac{SS_{\text{between}}}{k-1}$	$F = \dfrac{MS_{\text{between}}}{MS_{\text{within}}}$
Within groups	SS_{within}	$N-k$	$MS_{\text{within}} = \dfrac{SS_{\text{within}}}{N-k}$	
Total	SS_{total}	$N-1$	–	–

Warning: What ANOVA does not say

A significant ANOVA result says that **not all means are equal**. It does not, by itself, tell you exactly which groups are different. For that, you need a planned comparison or a post-hoc test such as Tukey's HSD.

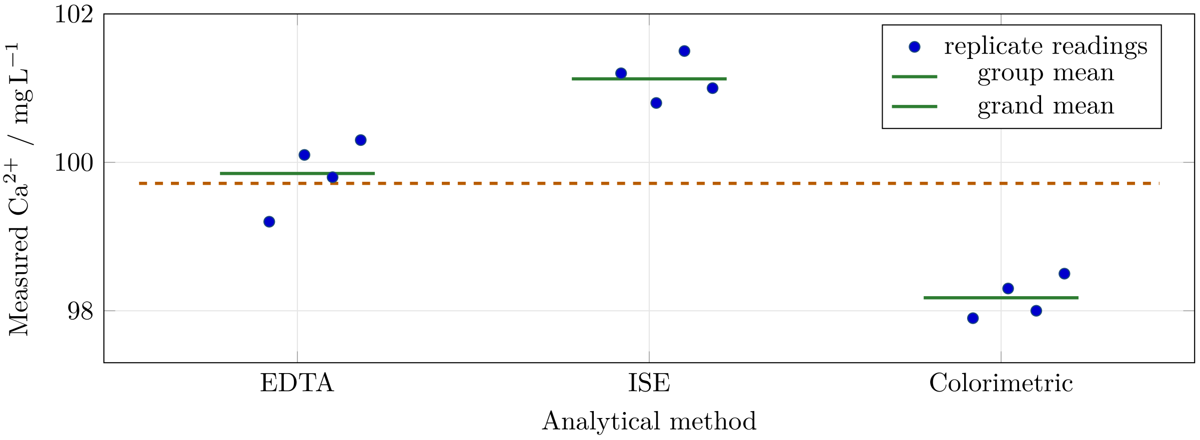
3 Worked example: comparing analytical methods

A laboratory class measures the calcium concentration in the same standard solution using three methods. The true value is close to $100\,\text{mg}\,\text{L}^{-1}$, but the immediate question is not accuracy against the true value. The ANOVA question is whether the methods give the same mean result.

Worked example: Calcium concentration by three methods

Method	Trial 1	Trial 2	Trial 3	Trial 4
EDTA titration	99.2	100.1	99.8	100.3
Ion-selective electrode	101.2	100.8	101.5	101.0
Colorimetric method	97.9	98.3	98.0	98.5

All values are in mg L^{-1}.



Step 1: calculate the group means and grand mean

$$\bar{x}_{\text{EDTA}} = 99.85, \qquad \bar{x}_{\text{ISE}} = 101.125, \qquad \bar{x}_{\text{Color}} = 98.175$$

$$\bar{x}_{\text{grand}} = 99.717$$

Step 2: calculate sums of squares

$$SS_{\text{between}} = \sum_i n_i(\bar{x}_i - \bar{x}_{\text{grand}})^2 = 17.512$$

$$SS_{\text{within}} = \sum_i \sum_j (x_{ij} - \bar{x}_i)^2 = 1.185$$

$$SS_{\text{total}} = 18.697$$

Step 3: complete the ANOVA table

There are $k = 3$ groups and $N = 12$ total measurements.

Source	SS	df	MS	F	Interpretation
Between methods	17.512	2	8.756	66.50	very large
Within methods	1.185	9	0.132		
Total	18.697	11	–	–	–

At the 5% significance level, F_{critical} for $\text{df}_1 = 2$ and $\text{df}_2 = 9$ is approximately 4.26. Since $66.50 > 4.26$, reject the null hypothesis.

Conclusion from the worked example

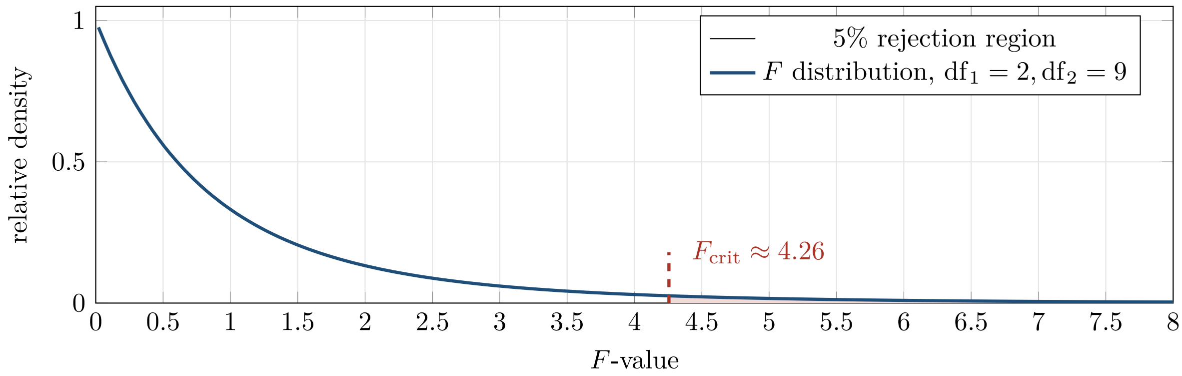
The three analytical methods do not give the same mean calcium concentration. The group means differ by more than would be expected from the scatter within each method.

4 Interpreting F, p, and effect size

The F-statistic is a ratio:

$$F = \frac{\text{variation between group means}}{\text{variation inside groups}}$$

A large F-value means the group means are separated relative to the natural scatter of repeated measurements.



p-value

The p-value is the probability of obtaining an F-value at least as extreme as the observed one, assuming the null hypothesis is true. A small p-value means the observed separation between means would be unlikely if all group means were really equal.

Warning: Significant is not automatically important

A statistically significant result may be chemically trivial if the difference is too small to matter in practice. A non-significant result may still be inconclusive if the sample size is too small or measurements are very noisy.

Effect size

A useful simple effect size for one-way ANOVA is:

$$\eta^2 = \frac{SS_{\text{between}}}{SS_{\text{total}}}$$

It estimates the fraction of total variation associated with group membership. In the worked example:

$$\eta^2 = \frac{17.512}{18.697} = 0.937$$

This means about 94% of the variation in the measurements is associated with the analytical method used. This does not prove mechanism, but it shows that the method effect is large in this dataset.

Quick exercise: Reading an ANOVA result

A one-way ANOVA comparing three indicators gives $F = 3.1$, $F_{\text{critical}} = 4.3$, and $p = 0.094$. What should be concluded at the 5% level?

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5 Assumptions and diagnostic plots

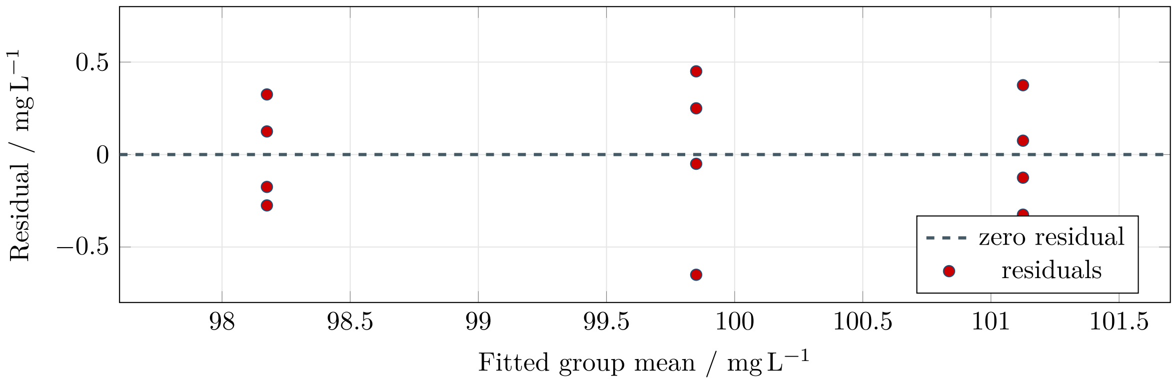
ANOVA is not just a formula. It has assumptions. In school chemistry, these assumptions should be treated as checks on whether the conclusion is defensible.

Assumption	Practical chemistry meaning
Independent measurements	One replicate should not mechanically determine the next one. Repeated readings from the same badly prepared stock solution are not fully independent evidence about preparation error.
Approximate normality	Residuals should not show extreme skew or obvious outlier behaviour. With small n, this is judged cautiously.
Similar variances	Groups should have roughly comparable scatter. A method with much larger scatter may require a different analysis or a careful limitation statement.
Comparable conditions	Same sample handling, temperature control, instrument calibration, and timing rules should apply across groups.

A **residual** is the difference between an observed value and its group mean:

$$e_{ij} = x_{ij} - \bar{x}_i$$

The residual plot below uses the worked example. The vertical axis shows how far each reading is from its own method mean.



How to read this diagnostic plot

For a simple classroom ANOVA, the residuals should look like ordinary scatter around zero. A pattern, a strong funnel shape, or one extreme point suggests that the ANOVA conclusion needs caution.

6 Post-hoc comparisons

If ANOVA is significant, you still need to find which groups differ. One common follow-up is Tukey's HSD test. In a balanced design, where every group has the same number of replicates, the simplified comparison threshold is:

$$\text{HSD} = q_{\alpha,k,N-k}\sqrt{\frac{MS_{\text{within}}}{n}}$$

For the worked example, $q \approx 3.95$, $n = 4$, and $MS_{\text{within}} = 0.132$.

$$\text{HSD} = 3.95\sqrt{\frac{0.132}{4}} = 0.72\ \text{mg L}^{-1}$$

Two method means differ significantly if their absolute difference is greater than $0.72\,\text{mg L}^{-1}$.

Pair	Difference in means / $mg\,L^{-1}$	Significant by HSD?
ISE vs EDTA	1.28	Yes
EDTA vs Colorimetric	1.68	Yes
ISE vs Colorimetric	2.95	Yes

Warning: Do not do post-hoc tests blindly

Post-hoc tests are for interpreting a significant ANOVA or for planned comparisons. They should be connected to the chemical question, not used as a fishing expedition.

7 Reporting ANOVA in chemical language

A statistical result should be written as a chemical conclusion, not as a disconnected number.

Basic reporting template

A one-way ANOVA was used to compare ____________ across ____________ groups. The effect of ____________ was / was not statistically significant, $F($____________, ____________$)$ = ____________, p = ____________. This suggests that ____________.

Example report

A one-way ANOVA was used to compare measured calcium concentration across three analytical methods. The effect of method was statistically significant, $F(2, 9) = 66.50$, $p < 0.001$. This suggests that the three methods do not give equivalent mean results for this calcium standard. A follow-up comparison showed that all three method means differed by more than the Tukey HSD threshold.

Quick exercise: Improve the conclusion

A student writes: “The p-value is small, so the experiment worked.” Rewrite this as a defensible chemical conclusion.

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8 Quick exercises

Quick exercise: Exercise 1: chloride in three water sources

A class measures chloride concentration in three water sources using the same titration procedure. All values are in $\mathrm{mg\,L^{-1}}$.

Source	Trial 1	Trial 2	Trial 3
A	24.8	25.1	24.9
B	25.4	25.6	25.3
C	26.1	25.9	26.2

a) Calculate the three group means.

b) Calculate the grand mean.

c) Complete the ANOVA table.

d) Compare F with $F_{\text{critical}} = 5.14$ at the 5% level.

e) Write a chemical conclusion.

Source	SS	df	MS	F
Between sources	______	______	______	______
Within sources	______	______	______	
Total	______	______	–	–

Quick exercise: Exercise 2: catalyst yield

A reaction is tested using three catalysts. Percentage yield is measured in four independent trials.

Catalyst	Trial 1	Trial 2	Trial 3	Trial 4
A	76.1	77.4	76.9	77.0
B	77.2	78.0	77.8	76.9
C	75.8	76.3	77.1	76.5

A one-way ANOVA gives:

$$SS_{\text{between}} = 2.232, \qquad SS_{\text{within}} = 2.545, \qquad F = 3.946, \qquad p = 0.0588$$

At the 5% significance level, $F_{\text{critical}} = 4.26$.

a) Should the null hypothesis be rejected?

b) What does the p-value suggest?

c) Why would it be wrong to claim that catalyst B is definitely best?

d) What additional evidence would help?

Quick exercise: Exercise 3: assumptions

For each situation, identify the assumption problem.

a) One group was measured on Monday using a freshly calibrated pH meter, while another was measured on Friday after the probe had dried out.

b) A student records five absorbance readings from the same cuvette without preparing independent samples.

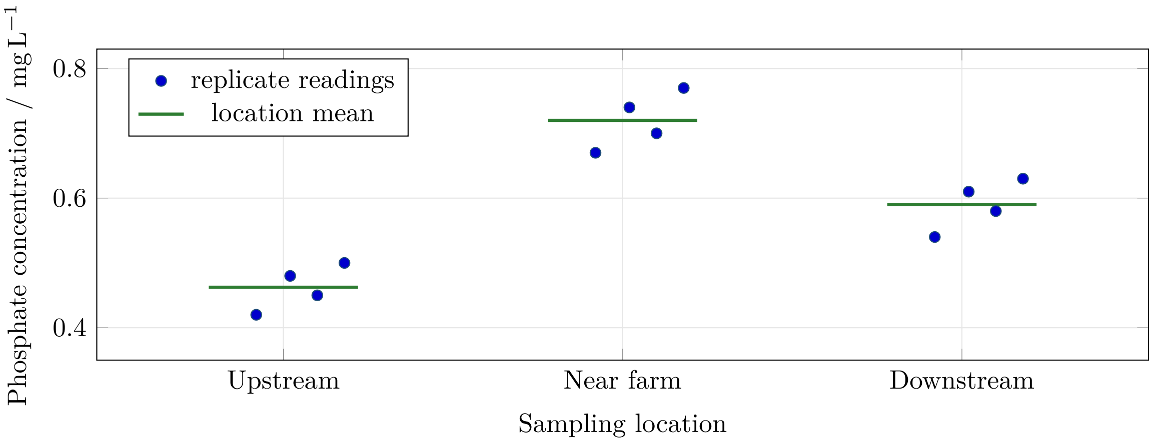
c) One catalyst group has yields from 60% to 95%, while the other groups vary only by about 2%.

d) One result is obviously impossible because the recorded titre is larger than the burette capacity.

9 Mini-practical: phosphate near a farm runoff site

A student group measures phosphate concentration in water samples from three locations: upstream, near a farm runoff point, and downstream. All values are in $\mathrm{mg\,L^{-1}}$.

Location	Trial 1	Trial 2	Trial 3	Trial 4
Upstream	0.42	0.48	0.45	0.50
Near farm runoff	0.67	0.74	0.70	0.77
Downstream	0.54	0.61	0.58	0.63



a) State the null and alternative hypotheses.

b) Calculate the group means and grand mean.

c) Complete the ANOVA table.

d) At the 5% level, $F_{\text{critical}} = 4.26$. Decide whether location affects phosphate concentration.

e) Calculate η^2.

f) Tukey's HSD for this dataset is approximately 0.078 $\mathrm{mg\,L^{-1}}$. Which pairs of locations differ?

g) Write a final chemical conclusion, including one limitation.

Source	SS	df	MS	F
Between locations	______	______	______	______
Within locations	______	______	______	
Total	______	______	–	–

10 Answer key

Choosing the method

a) ANOVA is appropriate: three juice brands.

b) One-sample t-test is more appropriate: one mean compared with a certified value.

c) Two-sample t-test is appropriate if there are only two indicators.

d) ANOVA is appropriate: four catalysts.

Reading an ANOVA result

Since $F = 3.1 < 4.3$ and $p = 0.094 > 0.05$, do not reject the null hypothesis at the 5% level. The data do not provide strong enough evidence that the three indicator means differ. This does not prove the indicators are identical; it means the evidence is insufficient under this experiment.

Improve the conclusion

Acceptable answer: "The ANOVA gives statistical evidence that the group means are not all equal. The chemical meaning depends on what the groups were. The conclusion should state which variable was tested and whether the observed difference is chemically meaningful."

Exercise 1

Group means:

$$\bar{x}_A = 24.93, \qquad \bar{x}_B = 25.43, \qquad \bar{x}_C = 26.07$$

Grand mean:

$$\bar{x}_{\text{grand}} = 25.48$$

Source	SS	df	MS	F
Between sources	1.936	2	0.968	41.48
Within sources	0.140	6	0.0233	
Total	2.076	8	–	–

Since $F = 41.48 > 5.14$, reject the null hypothesis. The three water sources do not have the same mean chloride concentration under this measurement procedure.

Exercise 2

Do not reject the null hypothesis at the 5% level because $F = 3.946 < 4.26$ and $p = 0.0588 > 0.05$. The result is close to the threshold, so it may justify further investigation, but it is not statistically significant at 5%. It would be wrong to claim catalyst B is definitely best because ANOVA has not shown a significant catalyst effect, and the observed differences may be due to experimental scatter. More independent trials, better control of temperature and purity, and possibly planned comparisons would help.

Exercise 3

a) Comparable conditions / systematic difference between groups.

b) Independence problem; repeated readings are not the same as independent sample preparations.

c) Similar variance assumption problem.

d) Data recording or procedural error; should be investigated before analysis.

Mini-practical

Hypotheses:

$$H_0 : \mu_{\text{upstream}} = \mu_{\text{near}} = \mu_{\text{downstream}}$$

$$H_A : \text{at least one location mean is different}$$

Group means:

$$0.4625, \qquad 0.7200, \qquad 0.5900 \text{ mg L}^{-1}$$

Grand mean:

$$0.5908 \text{ mg L}^{-1}$$

Source	SS	df	MS	F
Between locations	0.13262	2	0.06631	42.40
Within locations	0.01408	9	0.00156	
Total	0.14669	11	–	–

Since $F = 42.40 > 4.26$, reject H_0. Location has a statistically significant effect on phosphate concentration in this dataset.

$$\eta^2 = \frac{0.13262}{0.14669} = 0.904$$

About 90% of the variation in this dataset is associated with location. With HSD $= 0.078$ mg L^{-1}, all pairwise differences are significant:

$$0.7200 - 0.4625 = 0.2575, \quad 0.5900 - 0.4625 = 0.1275, \quad 0.7200 - 0.5900 = 0.1300$$

A defensible conclusion: phosphate concentration was lowest upstream, highest near the farm runoff point, and intermediate downstream. ANOVA showed a significant effect of location. A limitation is that the data represent a small number of samples and do not prove the farm is the only cause; time of sampling, rainfall, flow rate, and other nutrient sources should be controlled or recorded.